Rishik Sood

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EXPERIENCE

DataReadyAI July 2026 - current

AI Engineer SYD, AU

- Building a **language-agnostic data lineage system** using **Python** and **React-flow**. Given a code base, it is capable of tracking column transformations from raw to gold layers and build analysis on top of it.
- Developing **Ontologies** from customer structured and unstructured data and consuming them via **Cortex** agents.

Amazon Jan 2022 – Jan 2023

Software Development Engineer BLR, IN

- Built the frontend for internal debugging platform **Sherlock** end-to-end with **React, Redux, and REST APIs**, delivering a fast, usable interface for log analysis and issue-triaging across distributed services.
- Debugged and resolved **compliance and security risks** across two Tier-1 production services spanning 13+ teams, reasoning across systems owned by others.
- Built and maintained **CI/CD pipelines** with **Jenkins** and practiced **TDD** to support stable production deployments in an **Agile** delivery cycle.

University of Technology, Sydney Nov 2023 – Present

Casual Academic SYD, AU

- Delivered tutorials and lab sessions for **Programming 2 (Java/Python)** and **Data Structures & Algorithms (C++)**.
- Built **AI-assisted workflow tooling** with **Claude** and prompt engineering to automate feedback generation, streamlining the marking process for large cohorts.

PROJECTS

MemHateCaptioning

- Developed a multimodal framework for **hate speech detection in memes** using context-aware caption generation.
- Built a **retrieval-augmented (RAG)** pipeline integrating **CLIPCap, EasyOCR, LLMs (GPT, T5)** and web-context retrieval to generate explanatory captions for meme interpretation.
- Published and presented a research paper in WWCC'25 Conference.

Blunder Psychologist

- Architected a multi-user chess analysis platform end-to-end: a **C++** engine, a **Python/FastAPI** backend with a **LangGraph** agent layer, and a **TypeScript/React** frontend.
- Built a per-user **RAG** pipeline with **FAISS** retrieval and **LLM**-generated annotations grounded in each player's own game history.
- Built test-first with a **CI** suite and **Docker**, backed by a **PostgreSQL** job queue for asynchronous processing and realtime status streaming to the UI.

EDUCATION

University of Technology, Sydney Aug 2023 – Aug 2025

Master's of Information Technology (Data Analytics) Sydney, Australia

- **WAM**: 87.5%

SKILLS

Languages : TypeScript, JavaScript, Python, Node.js, Java, C++, SQL

Frontend : React, Next.js, Redux, REST API integration, reusable component design

Backend/Platform : Node.js, FastAPI, REST APIs, WebSockets/SSE, GitHub API, Docker, AWS, CI/CD, PostgreSQL

AI/ML : LLMs, RAG, LangGraph, AI-assisted workflows, prompt engineering, FAISS

ACHIEVEMENTS

Codeforces Candidate Master (1991) | **CodeChef** 5⋆ (2033), Advanced DSA Certification | **AtCoder** 4 Kyu (1277) | **UTS Programming Contest** 1st place (2x) | **ICPC** Regionals 2021: 110/550 | **INOI** 2018 Qualifier | **Chess** 1650 FIDE